SC

SEAT No.________

[588 A-46]

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SARDAR PATEL UNIVERSITY
B.C.A. (3rd Semester) (CBCS) Examination 2017
Thursday, November 16, 2017
02:00 pm to 05:00 pm
US03CBCA03 || ADVANCED DATA AND FILE STRUCTURE

Marks: 70

Q:1 Select an appropriate answer for the following. **[10]**

1. __________ means a link between parent and its child.
(a) Branch Degree (b) Height (c) Leaf (d) Degree

2. An array is a ____________ data structure.
(a) Heterogeneous (b) Non- Linear (c) Unordered (d) Homogeneous

3. If the range of index varies from L_______U then size of the array is____________.
(a) U / L – 1 (b) U – L + 1 (c) L – U + 1 (d) L / U – 1

4. A graph (or digraph) is termed as ___________ if all the edges in it are labeled with some weights.
(a) Label graph (b) Line graph (c) Weighted graph (d) Double graph

5. A node whose outdegree is 0 is called __________.
(a) Sink node (b) Source node (c) Self loop (d) Single node.

6. Maximum number of nodes possible in a binary tree of height h is __________.
(a) 2 ^ (h + 1) (b) 2 ^ (h - 1) (c) 2 ^ h + 1 (d) 2 ^ h - 1

7. The process of finding the data from its data structure is called __________
(a) Sorting (b) Searching (c) Deletion (d) None of these

8. ______________technique requires an ordered table to search a particular record in the table.
(a) Sequential search (b) Sorting (c) Binary search (d) None of these

9. The lowest level of index is ______________.
(a) Track index (b) Prime index (c) Index area (d) Master index

10. Record is also known as group or ________
(a) Item (b) Segment (c) Entity (d) None of these

Q:2 **Answer the following questions. (Attempt any ten)** **[20]**

1. Write the formula for address calculation of 1-D array element and explain it.
2. Define root and leaf of a tree.
3. List 2 applications of an array.
4. What is loop and cycle of a Graph?
5. Draw the Binary Tree for (A-B)+c*(E/F).
6. Define Directed and Undirected Graph.
7. List the applications of sorting.
8. Define Sequential Search.
9. List the applications of Searching.
10. What do you mean by Transaction? Which types of transactions are performed on the file?
11. Define: File, Database.
12. Write down the syntax and purpose of open statement for Input mode.

Q:3 A. Define array. Explain 1-D array with declaration and initialization. **[06]**
B. Explain Sparse Matrix in detail. **[04]**

OR

Q:3 A. List the representation of 2-D array in the memory. **[06]**
B. Explain applications of tree. **[04]**

Q:4 A. Explain the representation of Binary tree. **[06]**
B. Draw the binary tree for following expressions: **[04]**

Inorder	D	B	H	E	A	I	F	J	C	G
Preorder	A	B	D	E	H	C	F	I	J	G

OR

Q:4 A. Explain deletion of a node from binary tree (all 4 conditions) with example. **[06]**
B. What are the types of traversal of Binary tree? Explain any two with an example. **[04]**

Q:5 What is searching? List and explain searching techniques with algorithms. **[10]**

OR

Q:5 What is sorting? List sorting techniques and explain any one of them with algorithm. **[10]**

Q:6 A. Write a short note on Multiple buffering. **[05]**
B. Write a detail note on processing of Direct file. **[05]**

OR

Q:6 A. Explain the structure of index sequential file supported by IBM. **[05]**
B. List and explain different types of record layout supported by sequential file. **[05]**

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